Benjamin Li

25benjaminli@gmail.com | +1-973-757-4350 | benjaminli.net | github.com/25benjaminli

Education

Cornell University — Ithaca, NY
B.A. in Computer Science, Minor in Mathematics; GPA: 3.96 — *Expected Dec 2028*

Experience

Cornell Computational Imaging Lab — Ithaca, NY
Undergraduate Researcher — *Oct 2025 – Present*

- Developing machine learning algorithms for rapid, interpretable microscopy imaging under variable acquisition conditions.
- Designing transformer variants and uncertainty-calibration methods using conformal prediction, Bayesian inference, and evidential learning; contributed to quantile-conditioned U-Net and led research on safer, reduced photon-exposure imaging via uncertainty-aware denoising (both under review NeurIPS 2026).

Regeneron Pharmaceuticals — Tarrytown, NY
Software Engineering Intern — *Jun 2025 – Aug 2025*

- Built and open-sourced quantum optimization software for protein–ligand docking, formulating docking as a max-weight clique problem and converting it to quadratic unconstrained binary optimization.
- Implemented QAOA-based solvers, benchmarked them against classical brute-force approaches, and presented results to Regeneron and IBM research teams.

New Jersey Academy of Sciences — Remote
Software Engineering Intern — *Mar 2025 – May 2025*

- Designed and operated a judging system that processed results for approximately 200 students and 80 judges at the NJAS research symposium, continued usage through 2026.
- Replaced manual result verification with a reusable, open-source workflow featuring a comprehensive REST API and cloud firestore caching used by over a dozen volunteers and event organizers.

Montclair State University — Montclair, NJ
Research Intern — *Jan 2023 – Nov 2023*

- Developed computer vision framework for detecting toxin-producing cyanobacteria, improving on SOTA methods by more than 10% (avg. precision); attended IEEE conference and paper published in proceedings.
- Handled data collection, uniting local data from dept. of environmental science with online iNaturalist data; various methodological improvements, e.g. genetic hyperparameter optimization, specialized data augmentations.

Inventurn — Remote
Founder & Core Backend Developer — *May 2022 – Nov 2024*

- Built applications for business and nonprofits, including an outreach platform serving thousands of New Jersey volunteers.
- Focused on APIs and database logic; also gained experience with web3 and blockchain infrastructure.

Projects

Evidential Learning for Image Reconstruction

- Implemented novel uncertainty quantification via evidential deep learning for reconstruction; wrote related article featured as editor's pick for Towards Data Science, publication with 150,000+ monthly subscribers.

Autograd from Scratch

- Led a four-person team in building a vectorized reverse-mode automatic differentiation engine with matrix operations, common nonlinearities, SGD, and computation-graph visualization; validated it at 98% MNIST test accuracy.

Learning-Based Cache Eviction

- Built synthetic workloads and a data pipeline for an ML eviction policy that improved cache hit rate by up to 8% over the least-recently-used baseline.

Selected Honors & Publications

Honors: Regeneron Science Talent Search Top 40 Finalist (2025); 1st Place Math and CS, NJ Academy of Sciences Symposium (2024); IEEE UEMCON Best Paper, AI/ML (2023)
Publications: First author, *Journal of Medical Imaging* (2025); first author, *IEEE UEMCON* (2023)

Technical Skills

Languages: Python, Java, JavaScript, C/C++, SQL, LaTeX
Frameworks/Libraries: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, Pandas, OpenCV, Qiskit
Tools: Git, Docker, Linux/Unix, Slurm